RODOS: Hands on

Rishav

May 17, 2021

RODOS: Recap

Features

1. Hardware abstraction
2. Timing abstraction
3. Preemptive dynamic schedule

Philosophy

1. Dependability*
2. Simplicity

* *Dependability* is the ability to provide services that can defensibly be trusted within a time-period.

Goals

1. Create a *thread*
2. Abstractions: *time* and *hardware*
3. Run two threads with *different priorities*
4. *Communication* between different threads
5. *Tele-commands*
6. *Remote access* of Discovery board

C++ Intro

RODOS
Hello World
from Scratch

Hello World !!

ध्यान दिनुपर्ने कुराहरु !!

```
1   // The class Thread is extended by a custom run() procedure, which
2   // writes Hello World to the UART with PRINTF().
3
4   #include "rodos.h"
5
6   class HelloWorld : public StaticThread<>
7   {
8       void run()
9       {
10          PRINTF("Hello World!\n");
11      }
12  } hello_world;
```

Threads (Tasks)

```cpp
// The thread LowPriorityThread writes the character "." every second
// and is interrupted every five second by the thread HighPriorityThread,
// which writes the character "*".

#include <rodos.h>

class HighPriorityThread : public StaticThread<>
{
public:
    HighPriorityThread() : StaticThread("HiPriority", 55) {}
    void run()
    {
        while (1)
        {
            suspendCallerUntil(NOW() + 5 * SECONDS);
            PRINTF("*\n");
        }
    }
} highprio;

class LowPriorityThread : public StaticThread<>
{
public:
    LowPriorityThread() : StaticThread("LowPriority", 25) {}
    void run()
    {
        while (1)
        {
            suspendCallerUntil(NOW() + 1 * SECONDS);
            PRINTF(".\n");
        }
    }
} lowprio;
```

Communication between threads

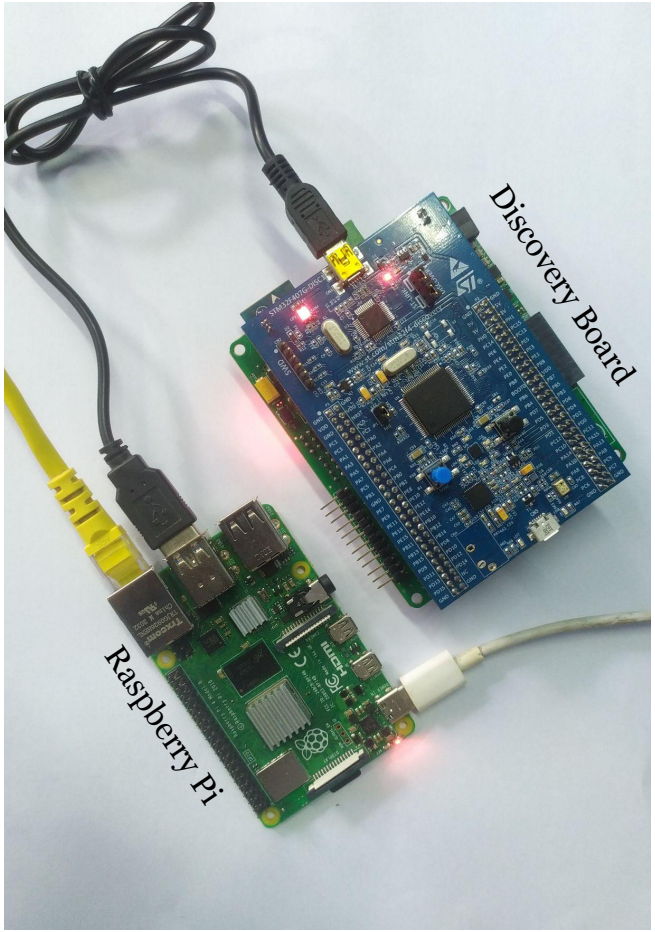
```cpp
// The Publisher-Thread post every one second an ascending counter value,
// while the Subscriber-Thread simply displays the received integer value.

#include <rodos.h>

Topic<long> counter1(-1, "counter1");

class MyPublisher : public StaticThread<>
{
public:
    MyPublisher() : StaticThread("SenderSimple") {}
    void run()
    {
        long cnt = 0;
        TIME_LOOP(1 * SECONDS, 1 * SECONDS)
        {
            PRINTF("Published: %ld\n", ++cnt);
            counter1.publish(cnt);
        }
    }
} publisher;

class MySubscriber : public SubscriberReceiver<long>
{
public:
    MySubscriber() : SubscriberReceiver<long>(counter1) {}
    void put(long &data)
    {
        PRINTF("Received: %ld\n\n", data);
    }
} subscriber;
```

Remote access to OBC

Your computer

Internet

SSH using PuTTy



My home